

LangGraph vs LangChain

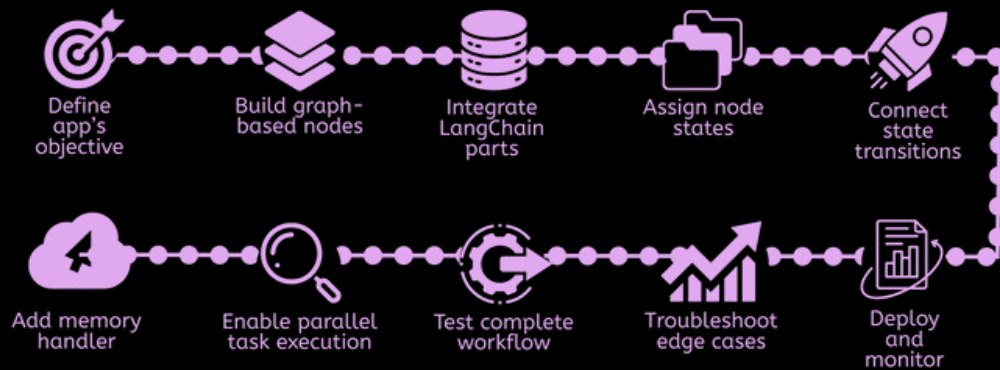


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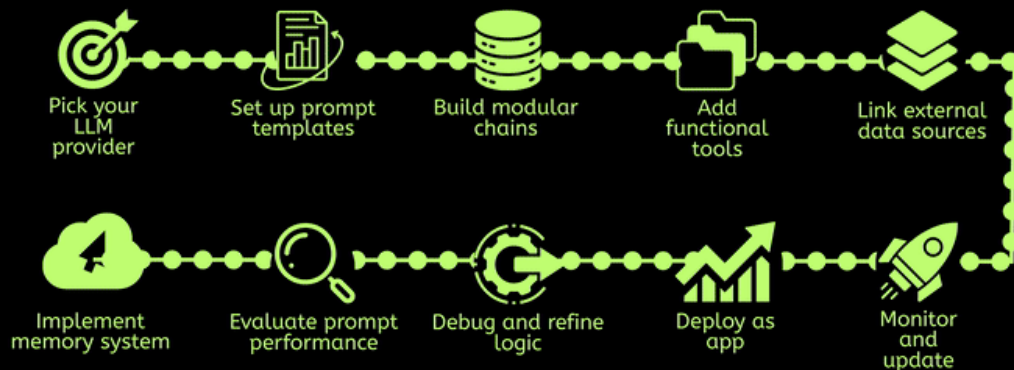
LANGGRAPH

A graph-driven framework for building dynamic, multi-agent LLM applications.



LANGCHAIN

A developer-oriented framework for creating intelligent LLM apps using tools, memory, and modular chains.



LANGGRAPH (step-by-step) LangGraph is a graph-driven framework for building dynamic, multi-agent AI workflows.

1. **Define app objective** – Clearly state what your app should achieve.
2. **Build graph-based nodes** – Divide the workflow into nodes, each handling a specific function.
3. **Integrate LangChain parts** – Use LangChain components (tools, prompts, retrievers) inside these nodes.
4. **Assign node states** – Give each node a status like *active*, *waiting*, or *complete* to track progress.
5. **Connect state transitions** – Define how one node leads to another based on outcomes or triggers.
6. **Deploy and monitor** – Launch the app and keep track of performance, uptime, and user behavior.
7. **Troubleshoot edge cases** – Identify rare or confusing user inputs and handle them gracefully.
8. **Test complete workflow** – Run the entire graph end-to-end to ensure smooth communication between nodes.
9. **Enable parallel task execution** – Allow multiple nodes to run simultaneously for faster results.
10. **Add memory handler** – Integrate memory so the app remembers previous interactions or states.

LANGCHAIN (step-by-step) LangChain is a developer-focused framework for creating modular, tool-powered LLM applications.

1. **Pick your LLM provider** – Choose the base model (like OpenAI, Anthropic, or Gemini).
2. **Set up prompt templates** – Design reusable prompt formats for consistent LLM responses.
3. **Build modular chains** – Connect multiple prompts and tools to form a logical pipeline.
4. **Add functional tools** – Attach external tools like search APIs or calculators.

5. **Link external data sources** – Connect databases, PDFs, or APIs to provide context-rich information.

6. **Monitor and update** – Regularly check performance and make updates to prompts or logic.

7. **Deploy as app** – Turn the workflow into a production-ready application.

8. **Debug and refine logic** – Fix errors, optimize chains, and refine responses through testing.

9. **Evaluate prompt performance** – Measure how accurately prompts generate desired outputs.

10. **Implement memory system** – Add short-term or long-term memory for contextual continuity.

In short:

- **LangGraph** builds **dynamic, multi-agent AI flows**.
- **LangChain** builds **structured, tool-based LLM apps**.